

Dhruv Madhwal

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EDUCATION

Arizona State University, Tempe, AZ

Master of Science, Computer Science

Sep 2024 - May 2026

GPA: 4.11/4

Relevant Courses: Statistical Machine Learning, Artificial Intelligence, Topics in NLP, Data Intensive Systems for Machine Learning

Birla Institute of Technology and Science - Pilani, India

Master of Science, Physics & Bachelor of Engineering, Electronics and Instrumentation Engineering

Aug 2017 - May 2022

Relevant Courses: Machine Learning, Foundations of Data Science, Probability and Statistics

TECHNICAL SKILLS

- **Languages:** Python, SQL, C/C++, MATLAB
- **ML / DL:** scikit-learn, XGBoost, LightGBM, PyTorch, TensorFlow/Keras, Hugging Face Transformers, OpenCV, Tesseract
- **LLM + Retrieval (RAG):** LangChain, LangGraph, AutoGen, AWS Bedrock, FAISS, Pinecone, ChromaDB
- **Data Platforms & Streaming:** Spark, Kafka, PostgreSQL, MySQL, MongoDB
- **MLOps & Serving:** FastAPI, Flask, MLflow (tracking/registry), Docker, Git, CI/CD, AWS (S3, Lambda)

RESEARCH EXPERIENCE

Graduate Researcher | CoRAL Research Lab at ASU

Feb 2025 - Present

- **LLM Reliability via Prompt Disagreement:** Showed that disagreement between direct and decomposed prompting exposes knowledge gaps, and introduced DBA to abstain on unreliable multi-hop QA answers. ([Accepted at ACL 2026](#) )
- **Improving Compositional Reasoning in CLIP VLMs:** Showed lexical concreteness improves hard-negative quality, and paired it with a concreteness-aware loss to push CLIP to SOTA compositional reasoning. ([Under review at COLM 2026](#) )

WORK EXPERIENCE

Data Scientist | Trukker Technologies

July 2022 - July 2024

- Performed customer segmentation on 11000+ shippers using PCA and K-Means clustering on RFM metrics and lane diversity; identified 4 high-value segments driving 8% month-over-month increase in repeat bookings.
- Built Spark ETL pipelines to standardize master data from multiple source systems, applying business logic and normalization to publish MongoDB source-of-truth records. Cut duplicates by 62%, and streamline data for Ops, onboarding, and BI.
- Implemented event-driven Kafka order-to-invoice sync, consuming events and leveraging MongoDB Change Streams with idempotency, retries, and DLQ to improve throughput from 900 to 1,100 events/min and reduce invoice mismatches 38%.

Data Science Intern | Carelon Global Solutions

Jan 2022 - June 2022

- Architected a document-classification pipeline by fine-tuning BioBERT on prior-auth medical records, deployed as a Dockerized Flask API, achieved 96% classification accuracy.
- Built OCR pre-processing with OpenCV+Tesseract with 98% field-level accuracy. Emitted normalized JSON with PHI redaction and schema validation.

AI/ML Engineering Intern | Samsung Research Institute

July 2021 - December 2021

- Prototyped an on-device, context-aware recommendation system for Galaxy apps using user behavior patterns and Bixby signals. Achieved 71% Hit@5 accuracy with 78% model compression via post-training quantization with TFLite.
- Automated the ML lifecycle using MLflow. Set up versioned datasets, scheduled retraining, and ensured fully reproducible runs across experiments and model artifacts.

PROJECTS

Multi-Hop Reasoning Agent

- Built a multi-agent QA system with LangGraph and LangChain, integrating RAG, Tavily web search, and planner, retriever and executor agents for multi-hop factual question answering.
- Built a Streamlit interface to visualize end-to-end reasoning traces, state flow, tool calls, and code execution for interpretable, evidence-backed answers.

Machine Unlearning in Small Language Models

- Applied machine unlearning methods (gradient ascent, random labeling) to 3–4B parameter LLMs/SLMs, reducing overlap with target responses to 0.19 BLEU, 0.23 ROUGE-L, and 0.81 BERTScore on Wikipedia Person Unlearn.
- Evaluated 4 model variants for post-unlearning utility on TruthfulQA, with less than an 8-point drop in general performance.

DishCovery - Dish-Level Restaurant Search

- Built a dish-level restaurant search platform that translated natural language queries and dietary requirements into structured filters and NL-to-SQL queries to find matching dishes.
- Developed a scalable menu data pipeline using Amazon S3, S3-triggered AWS Lambda to automate VLM/LLM-based menu parsing, with JSON validation, FastAPI for serving search APIs, and Snowflake for structured, queryable data.